Preparation Guide

1. Core Tooling

🐳 Docker

Docker is the foundation for everything we do in Day 1.

- **Mac / Windows:** Install Docker Desktop.
  - *Windows Tip:* Ensure **WSL2** is enabled and Docker is configured to use the WSL2 backend.
  - *Mac Tip:* If you have an M1/M2/M3 chip, Docker Desktop for Apple Silicon is required.
- **Linux:** Install `docker-ce` and `docker-compose-plugin` via your package manager.
  - *Linux Tip:* Run `sudo usermod -aG docker $USER` and restart your session to run docker without `sudo`.

☸️ Kubernetes (Minikube)

We will use Minikube as our local cluster for Day 2.

1. **Install kubectl:** The Kubernetes command-line tool. (Instructions)
2. **Install Minikube:** Download here.
   - We recommend using the **docker driver** for consistency.

📦 Optional but Recommended

- **Helm:** Kubernetes package manager. (Install)
- **VS Code Extensions:**
  - `Docker` (by Microsoft)
  - `Kubernetes` (by Microsoft)
  - `YAML` (by Red Hat)

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2. Pre-pulling Docker Images

To save time and bandwidth during the training, please pull these images onto your machine in advance.

Run the following command in your terminal:

```
# Day 1 Essentials
docker pull hello-world
docker pull alpine:latest
docker pull ubuntu:latest
docker pull nginx:latest
docker pull busybox:latest

# Application Runtimes
docker pull node:20-alpine
docker pull eclipse-temurin:21-jdk-alpine
docker pull eclipse-temurin:21-jre-alpine

# Databases & Infrastructure
```

```
docker pull postgres:16-alpine
docker pull redis:7-alpine
```

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3. Environment Check

Run these commands to verify your setup:

```
# Check Docker
docker --version
docker run --rm hello-world

# Check Kubernetes (Minikube)
minikube version
minikube start --driver=docker
kubectl get nodes
minikube stop

# Check Tools
kubectl version --client
helm version
```

🚀 Troubleshooting

- **Network Issues:** If you are behind a corporate proxy, ensure Docker Desktop is configured with your proxy settings.
- **Virtualization:** Ensure Virtualization is enabled in your BIOS/UEFI settings (common for Windows/Linux users).
- **Disk Space:** Ensure you have at least **10GB** of free space for images and the Minikube virtual disk.

> *[!IMPORTANT] If `minikube start` fails, try `minikube delete` and then start again. If you're on a Mac with Apple Silicon, the `--driver=docker` is usually the most stable choice.*